

Urinary Tract Infections (UTIs) – Summary

Definitions:

- **Uncomplicated UTI:** UTI confined to the bladder (also known as cystitis) in afebrile women or men
No systemic symptoms (e.g. no fever)
- **Complicated UTI:** Infection beyond the bladder in women and men
 - Includes pyelonephritis, febrile or bacteremic UTI, catheter-associated UTI, prostatitis
 - Symptoms: flank pain, fever, other signs of systemic illness (e.g. hemodynamic instability)
- **Asymptomatic Bacteriuria:** Presence of bacteria in the urine of an *asymptomatic* individual

Most common causing-pathogens:

- **Uncomplicated UTI:** *E. coli* > *P. mirabilis*, *K. pneumoniae*, *S. saprophyticus*, *Enterococcus* spp.
- **Complicated UTI:** *E. coli* > *P. mirabilis*, *Enterococcus faecalis*, *Staphylococcus*, *Acinetobacter*

Diagnosis: Based on symptoms +/- urine culture and analysis

- **Urine culture indications:**
 - Always for complicated UTI (including pyelonephritis, prostatitis), pregnant individuals
 - Uncomplicated cystitis only if failure to respond to empiric therapy or early recurrence/relapse (<1 month) following therapy

Non-Pharmacological Treatment: Fluid hydration, Lactobacillus species probiotics, cranberry juice, surgery

Pharmacological Treatment

- **Asymptomatic bacteriuria:** ONLY treat for asymptomatic bacteriuria in pregnancy or if undergoing endo-urological procedures, treat according to culture sensitivities
- **Uncomplicated UTI (cystitis):**
 - **1st line:** Sulfamethoxazole-Trimethoprim (SMX-TMP) x 3 days **OR** Nitrofurantoin x 5 days **OR** Fosfomycin (1 dose)
 - **2nd line:** Fluoroquinolone x 3 days **OR** Beta-lactam x 3-7 days (avoid amoxicillin and ampicillin)
- **Complicated UTI:** Tx duration = 5-7 days unless condition is more severe
 - **With Sepsis:**
 - **1st line:** Fluoroquinolone (levofloxacin, ciprofloxacin), **OR** 3rd generation cephalosporins (e.g. ceftriaxone) **OR** 4th generation (cefepime) **OR** carbapenems **OR** piperacillin/tazobactam
 - **2nd line:** AMG (e.g. gentamicin) **OR** novel beta-lactam-beta lactamase inhibitors (e.g. ceftolozane-tazobactam)
 - **Without Sepsis:**
 - **1st line:** IV → Fluoroquinolone (levofloxacin, ciprofloxacin), **OR** 3rd generation cephalosporins (e.g. ceftriaxone) **OR** 4th generation (cefepime) **OR** piperacillin/tazobactam. **ORAL** → Fluoroquinolone (levofloxacin, ciprofloxacin) **OR** SMX-TMP
 - **2nd line:** IV → AMG **OR** Novel beta-lactam-beta-lactamase inhibitors (e.g. ceftolozane-tazobactam) **OR** Carbapenems. **ORAL** → Amoxi-clav **OR** cephalosporins
- **Recurrent UTI** (≥ 2 episodes of UTI in the past 6 months or ≥ 3 episodes in the past 12 months)
 - **Post-intercourse prophylaxis (post-coital):** SMX-TMP **OR** TMP **OR** Nitrofurantoin **OR** Cephalexin → 1 dose within 2 hours of intercourse
 - **Long-term low-dose prophylaxis:** SMX-TMP (1st line) **OR** TMP **OR** Nitrofurantoin **OR** cephalexin **OR** vaginal estrogens **OR** Fosfomycin (not often used)

Monitoring Parameters: Resolution within 3 days. If not, then reassessment by physician is recommended

Additional Key Points:

- **Pregnancy:** safe options include amoxicillin, amoxi/clav, cephalexin, ceftriaxone, fosfomycin
 - Avoid nitrofurantoin in > 36 weeks, SMX/TMP in 1st trimester and near term, FQs, aminoglycosides